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(54) **BIO-SIGNAL MEASURING DEVICE AND BIO-SIGNAL MEASURING METHOD**

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G16H 40/63 (2018.01)
H04L 67/12 (2022.01)
(52) **U.S. Cl.**
CPC **A61B 5/0002** (2013.01); **A61B 5/6898** (2013.01); **G16H 40/63** (2018.01); **H04L 67/12** (2013.01)

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See application file for complete search history.

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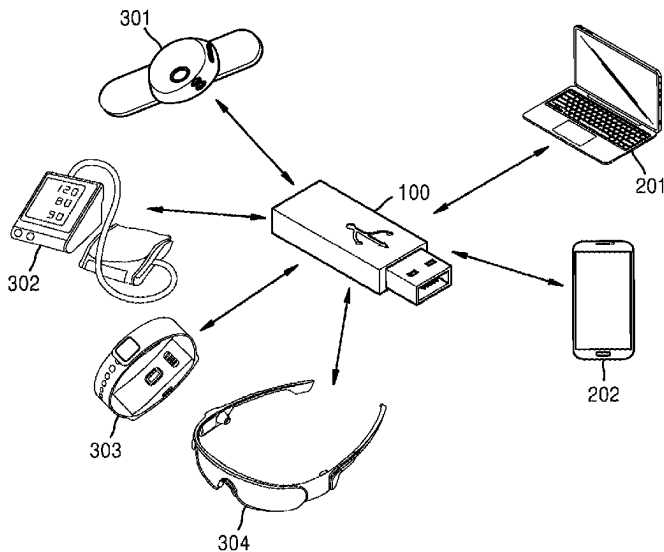
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(57) **ABSTRACT**

A bio-signal measuring device includes a signal detector detecting a bio-signal, a signal converter converting the bio-signal into a bio-digital signal, a memory, a communicator communicating with an external device, and a processor configured to operate in a first operation mode for storing the bio-digital signal in the memory or a second operation mode for transmitting the bio-digital signal to the external device. An electrocardiogram measuring system and method including a portable electronic device configured to facilitate transmitting an electrocardiogram digital signal from an electrocardiogram measuring device to an external device by pre-storing an identification of the external device, accessing a certificate stored in the external device, and authenticating the certificate of the external device, are also disclosed.

14 Claims, 7 Drawing Sheets



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FIG. 1

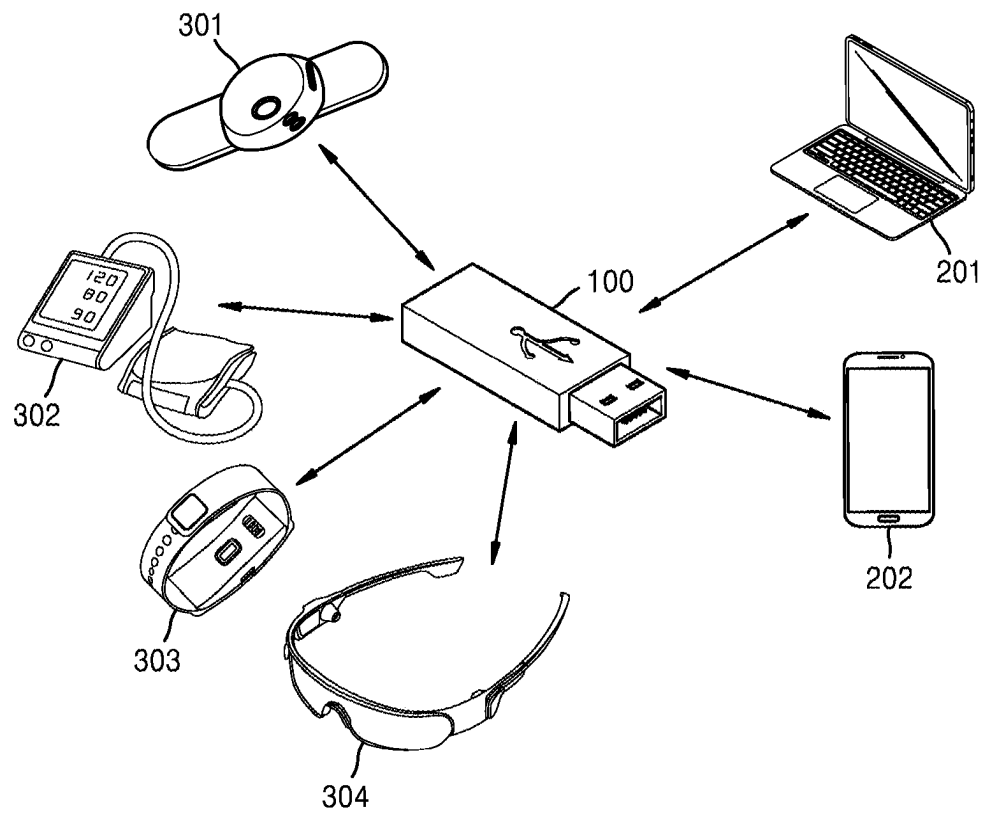


FIG. 2

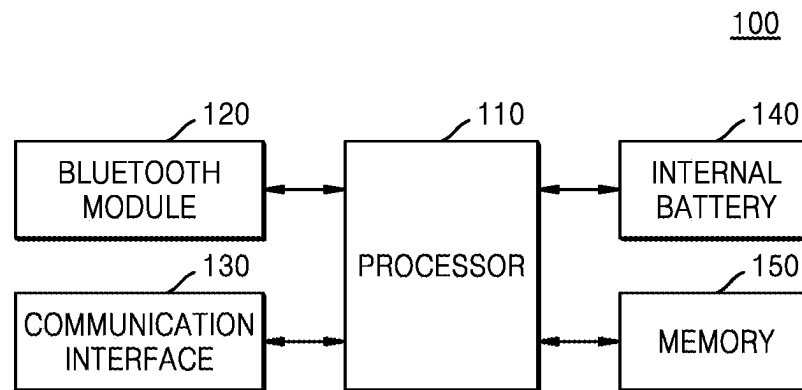


FIG. 3

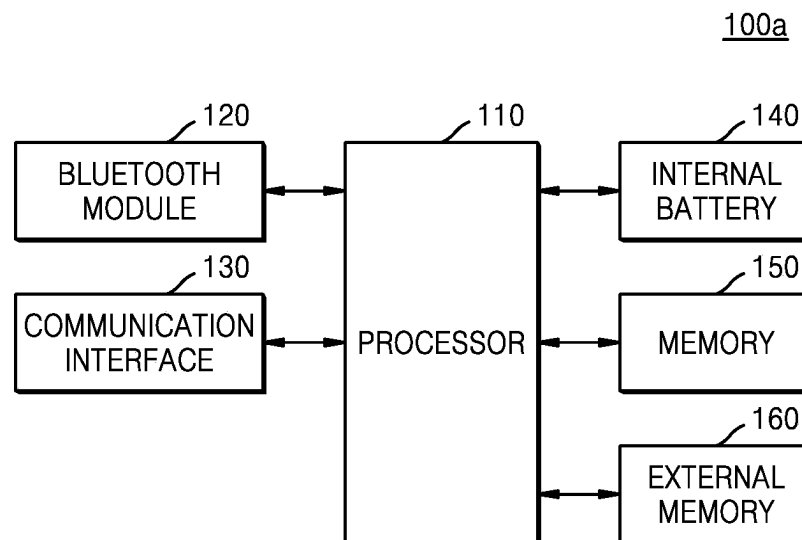


FIG. 4

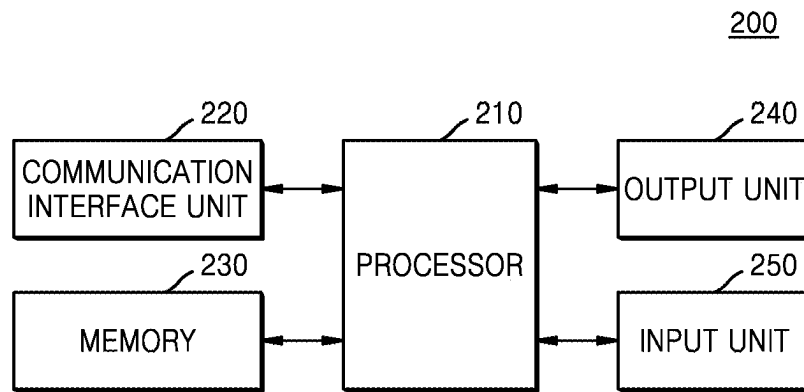


FIG. 5

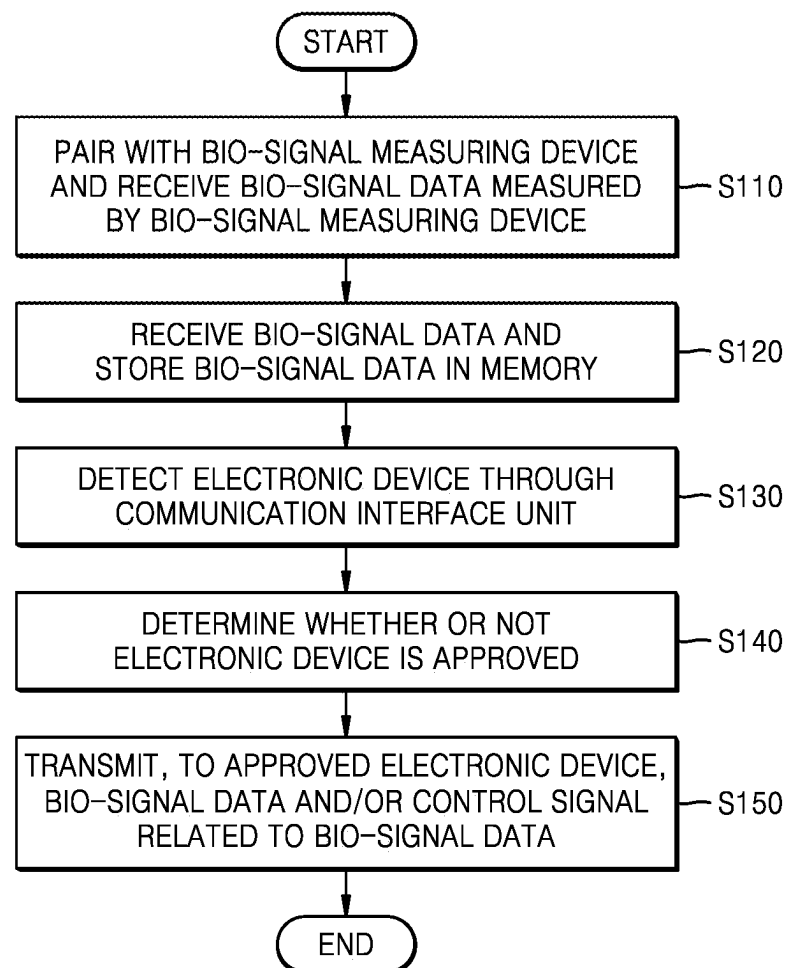


FIG. 6

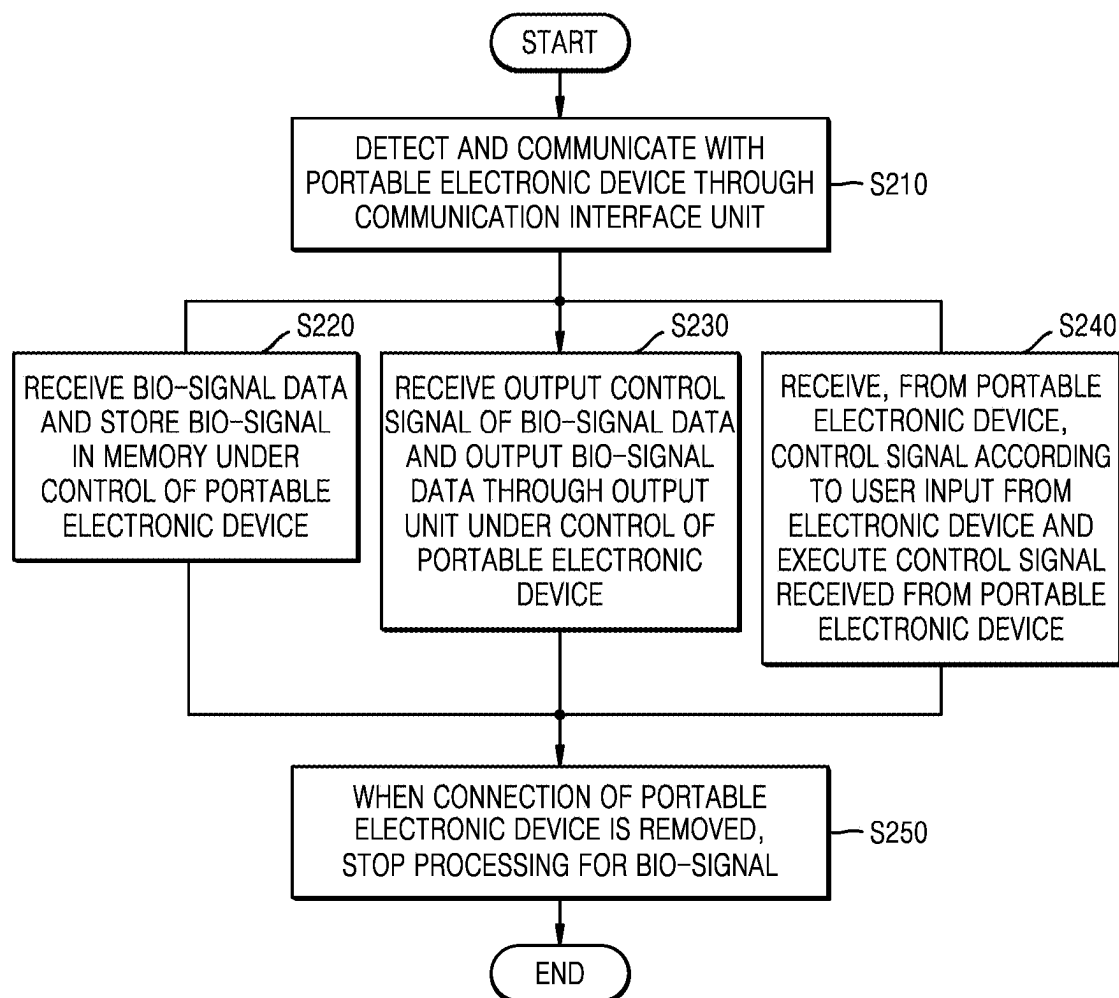


FIG. 7

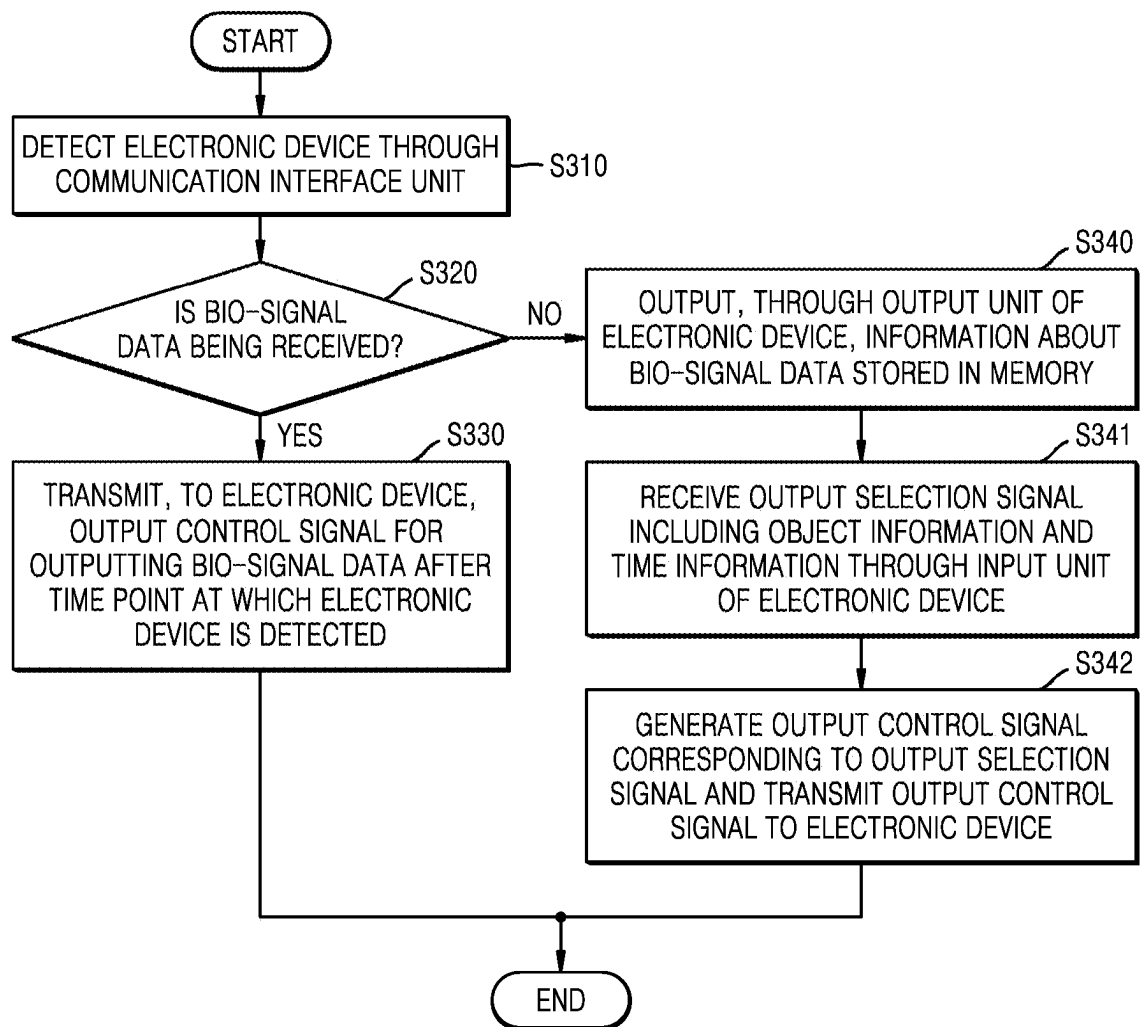


FIG. 8

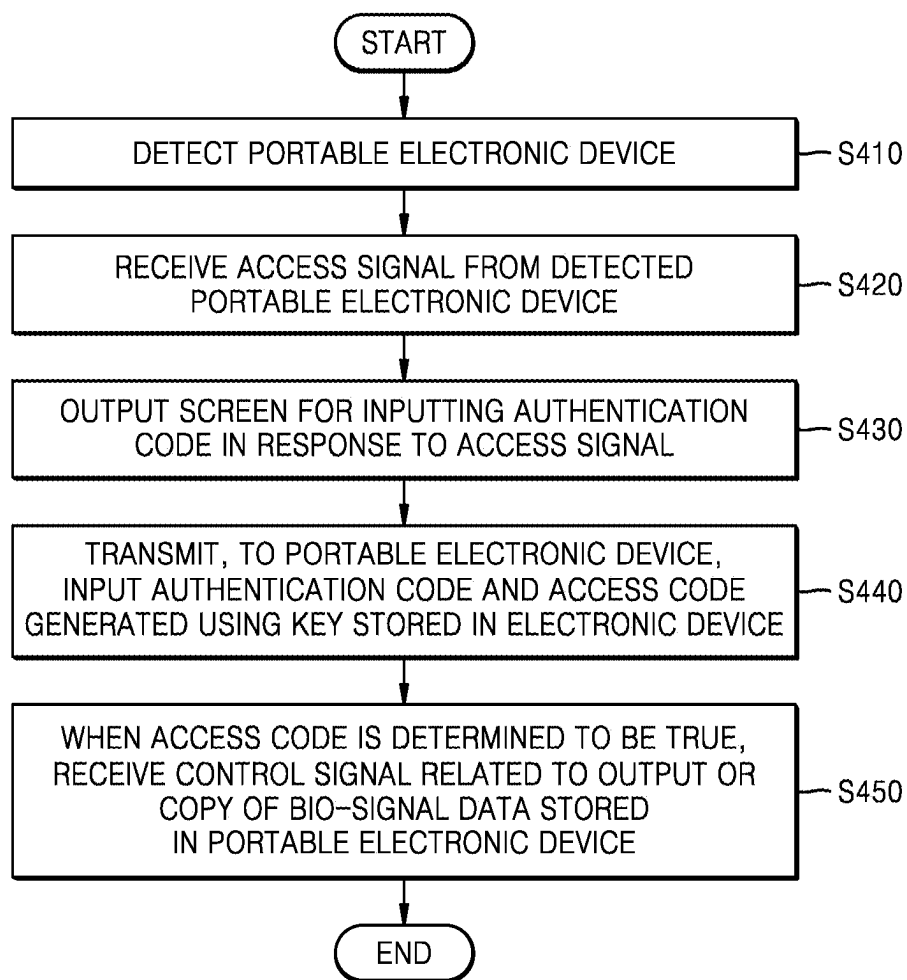
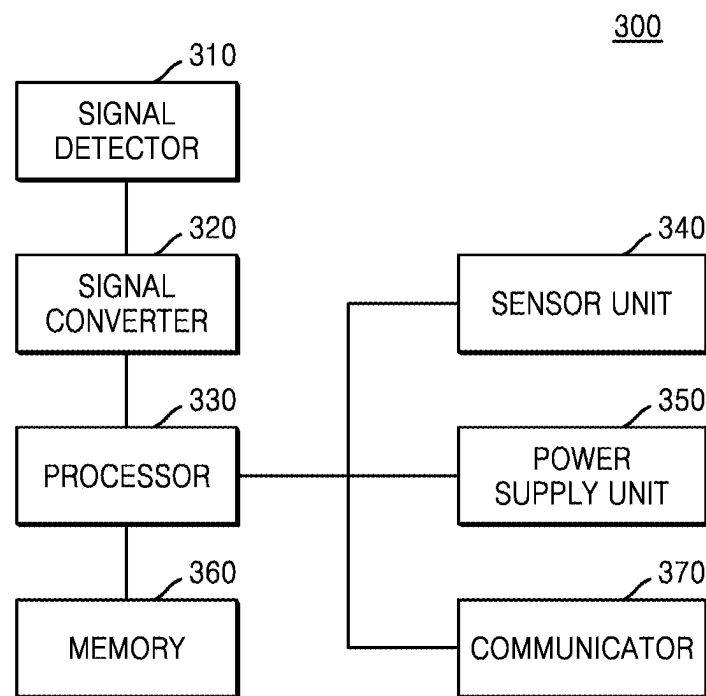


FIG. 9



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**BIO-SIGNAL MEASURING DEVICE AND
BIO-SIGNAL MEASURING METHOD****CROSS-REFERENCE TO RELATED
APPLICATION(S)**

This application is a continuation of U.S. patent application Ser. No. 17/036,082 filed Sep. 29, 2020, which issued as U.S. Pat. No. 12,029,524, by Jeong et al., entitled "BIO-SIGNAL MEASURING DEVICE AND BIO-SIGNAL MEASURING METHOD." All sections of the aforementioned application(s) are incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

One or more embodiments relate to a bio-signal measuring device and a bio-signal measuring method.

BACKGROUND

In order to maintain human life, there is a need for a process of enabling blood released by the heartbeat to flow along the arteries to all parts of the body without clogging and returning blood through the veins back to the heart. Accordingly, oxygen and nutrients may be supplied to the body's tissues, and consumed wastes may be removed through the metabolism.

However, when the human heart is in a poor condition, blood may not be properly delivered to particular parts of the body or a blood clot or embolism may occur in the blood. As a result, blood may become cloudy, and the cloudy blood may block capillaries, in particular tissues of the body, and cause tissue necrosis, and thus, the human life may be in danger. Therefore, in addition to clinical examinations, imaging tests and the like have been used to examine whether or not the heart is abnormal. Also, as an early diagnosis method, a method of determining whether or not a patient has an abnormality in the heart by measuring an electrocardiogram and displaying the measured electrocardiogram signal on a graph has also been widely used.

In other words, an electrocardiogram refers to recording of a potential change in the surface of the body that causes the mechanical activity of the heartbeat, such as contraction or expansion of the heart muscle. The electrocardiogram is a noninvasive test that is simple to measure, is easily reproduced, is easily repetitively recorded, and is inexpensive to test. The electrocardiogram has been used helpfully to diagnose arrhythmia and coronary artery disease (cardiac artery disease) and to monitor the progress of cardiac patients.

In general, the electrocardiogram is measured by attaching sensors for measuring electrocardiogram signals on the upper left and right and lower left and right of the chest and by using potential differences according to the location of the sensor.

SUMMARY

One or more embodiments include a portable electronic device for receiving bio-signal data from a measuring device and for storing and carrying the bio-signal data.

One or more embodiments include providing, through an output unit of an electronic device connected and communicating with a portable electronic device, bio-signal data stored in the portable electronic device or bio-signal data received through the portable electronic device.

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Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

A computer program according to one or more embodiments may be stored on a medium to execute, by using a computer, any one of methods according to one or more methods.

In addition, provided are other methods, other systems, and computer-readable recording media recording thereon a computer program for executing the method.

Other aspects, features, and advantages than those described above will become apparent from the following drawings, claims, and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

FIG. 1 is a view for explaining a bio-signal data processing system according to one or more embodiments.

FIG. 2 is a block diagram of a portable electronic device according to one or more embodiments.

FIG. 3 is a block diagram of a portable electronic device having an external memory according to one or more embodiments.

FIG. 4 is a block diagram of an electronic device according to one or more embodiments.

FIG. 5 is a flowchart showing a method of operation of a portable electronic device according to one or more embodiments.

FIG. 6 is a flowchart showing a method of operation of another portable electronic device according to one or more embodiments.

FIG. 7 is a flowchart showing operations between two portable electronic devices according to one or more embodiments.

FIG. 8 is a flowchart showing other operations between two portable electronic devices according to one or more embodiments.

FIG. 9 is a block diagram of a bio-signal measuring device according to one or more embodiments.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. Expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

While example embodiments are capable of various modifications and alternative forms, embodiments thereof are shown by way of example in the drawings and will herein be described in detail. The effects and features of the present disclosure and the accompanying methods thereof will become apparent from the following description of the embodiments, taken in conjunction with the accompanying

drawings. However, the present disclosure is not limited to the embodiments described below, and may be embodied in various modes.

Hereinafter, one or more embodiments will be described in detail with reference to the accompanying drawings.

FIG. 1 is a view for explaining a bio-signal data processing system according to one or more embodiments.

As shown in FIG. 1, in the bio-signal data processing system, bio-signal measuring devices **301**, **302**, **303**, and **304** (hereinafter referred to as a bio-signal measuring device **300**) store bio-signal data by measuring bio-signal data and transmitting the bio-signal data to a portable electronic device **100**.

The bio-signal measuring device **300** may be a device for measuring a bio-signal through the body of a user and measure blood pressure, blood sugar, heartrate, oxygen saturation level, sleep time, depression diagnosis, chemical analysis, weight, exercise load, an electrocardiogram, and the like.

The portable electronic device **100** stores, in a memory, the bio-signal data received from the bio-signal measuring device **300**. The portable electronic device **100** is connected to a communication port of electronic devices **201** and **202** (hereinafter referred to as an electronic device **200**) to perform various operations through the electronic device **200**. For example, the portable electronic device **100** may enable the bio-signal data to be output through the electronic device **200**. The portable electronic device **100** may transmit the bio-signal data to a preset electronic mail through the electronic device **200**. The portable electronic device **100** controls processing, such as preset output, data copy, and data movement, to be performed rather than stores the bio-signal data in the electronic device **200**.

The electronic device **200** may include one or more processors and a communication interface for communicating with external devices and may include an output unit with a user interface. As shown in FIG. 1, the electronic device **200** may be a desktop computer **201** or a smartphone **202** but is not limited thereto. The electronic device **200** may be implemented as a smartphone, a table personal computer (PC), a mobile phone, a video phone, an e-book reader, a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, a camera, a wearable device, or the like.

In the bio-signal data processing system, the bio-signal measuring device **300** refers to a device for measuring bio-signal data. For example, the bio-signal measuring device **300** may be a device which is mounted on an object noninvasively or invasively to detect an electrocardiogram. Here, the object may be a human, an animal, or part of the body of a human or animal such as the chest but is not limited thereto. The object may include all types of objects from which electrocardiograms may be detected or measured. Also, an electrocardiogram is a graph that records changes in a potential on the surface of the body causing the mechanical activity of the heartbeat such as contraction/expansion of the myocardium. Here, the meaning "detect an electrocardiogram" may mean the same as "detect a potential" on the surface of the body.

As shown in FIG. 1, the bio-signal measuring device **300** may be implemented in the form of a wristwatch, glasses, and the like to measure a bio-signal.

The bio-signal measuring device **300** may communicate with the portable electronic device **100** by including a near field communication module. Here, the near field communication module refers to a module for performing commu-

nication with an external device located at a short distance, according to near field communication methods such as Bluetooth, radio frequency identification (RFID), infrared data association (IrDA), ultra-wideband (UWB), and Zig-Bee. The bio-signal measuring device **300** may further include a band-type mounting part. The mounting part may be made of a flexible material that may be deformed to fit a curved surface of the surface of the body, for example, an elastic (i.e., flexible) fabric. The mounting part may be provided as a patch-type or a wearable type. As long as the bio-signal measuring device **300** is worn through the mounting part, the bio-signal measuring device **300** may detect a potential generated on the surface of the body by contacting the surface of the body of the object obj.

FIGS. 2 and 3 are block diagrams of portable electronic devices **100** and **100a** according to one or more embodiments.

As shown in FIG. 2, the portable electronic device **100** may include a processor **110**, a Bluetooth module **120**, a communication interface **130**, an internal battery **140**, and a memory **150**.

The processor **110** stores, in the memory **150**, bio-signal data received from the bio-signal measuring device **300**.

The processor **110** detects communication connection to the electronic device **200** through the communication interface **130**. The processor **110** may determine whether or not connection to the electronic device **200** is maintained by authenticating a certificate of the electronic device **200** or a password input from the electronic device **200**. For example, the processor **110** accesses the certificate stored in the electronic device **200** to determine whether or not a certificate is approved and maintains connection to the electronic device **200** storing the approved certificate. The processor **110** may determine whether or not the electronic device **200** is approved by transmitting, to the electronic device **200**, a control signal for providing a password input screen and receiving a password input through the password input screen.

The processor **110** may transmit, to the electronic device **200** that is connected, bio-signal data or an output control signal of the bio-signal data. For example, the output control signal controls received bio-signal data to be outputted. The output control signal may be changed according to a specification of an output unit of the connected electronic device **200** and may include programming to be dynamically changed according to a user input value from the electronic device **200**.

The processor **110** may transmit a stored signal of the bio-signal data to the connected electronic device **200**. The electronic device **200** stores the bio-signal data in the memory **150** in response to the stored signal.

The processor **110** may identify the electronic device **200** on the basis of an ID of the electronic device **200** and transmit a control signal to transmit the bio-signal data to a mail address corresponding to the electronic device **200**. The processor **110** may identify the validity of the ID of the electronic device **200** through an external server. The processor **110** may pre-store the ID of the electronic device **200** associated with the portable electronic device **100** and refer to the pre-stored ID of the electronic device **200**.

The electronic device **200** received the bio-signal data to a preset mail address in response to the control signal from the processor **110**. In another embodiment, the processor **110** may search for the ID of the electronic device **200** and store the bio-signal data in a storage area corresponding to the ID of the electronic device **200**.

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The Bluetooth module **120** receives the measured bio-signal data by communicating with the bio-signal measuring device **300**. The Bluetooth module **120** may be designed to communicate with the bio-signal measuring device **300** according to a time point at which the bio-signal measuring device **300** measures a bio-signal. The processor **110** stores, in the memory **150**, the bio-signal data received through the Bluetooth module **120**. The portable electronic device **100** may further include a state for notifying an operation of the Bluetooth module **120** or the like. The Bluetooth module **120** may receive and transmit data by using a frequency band of about 2400 MHz to about 2483.5 MHz. The Bluetooth module **120** may receive and transmit data while hopping one or more frequencies from among frequencies within a frequency band. For example, the Bluetooth module **120** may operate by receiving first data at a first frequency in one communication session and then changing the first frequency to a second frequency to receive second data.

The communication interface **130** may transmit the bio-signal data and the control signal to the electronic device **200**. The processor **110** stores, in the memory **150**, data received through the Bluetooth module **120** and transmits in real time, to the electronic device **200**, data received through the communication interface **130**. FIG. 1 illustrates that the communication interface **130** is a wired USB, but the communication interface **130** may be a wireless method.

The internal battery **140** supplies power to each component of the portable electronic device **100**. The internal battery **140** may supply power so that data is transmitted and received through the Bluetooth module **120** and/or the communication interface **130**. The internal battery **140** may be a lithium iron phosphate (LiFePO₄) battery but is not limited thereto and may be manufactured as a lithium cobalt oxide (LiCoO₂) battery, a lithium titanate battery, or the like. The internal battery **140** may be a rechargeable battery or a disposable battery. The internal battery **140** may be charged from an external power source device. The internal battery **140** may be charged by an external power source device through a communication link established through the communication interface **130**.

The memory **150** may store the bio-signal data measured through the bio-signal measuring device **300**. The bio-signal data is data generated by detecting a bio-signal of an object and refers to a signal having a frequency and an amplitude corresponding to a bio-signal measured in chronological order.

The memory **150** is hardware storing various types of data processed in the portable electronic device **100** and may store pieces of data processed by the processor **110** and pieces of data to be processed by the processor **110**. The memory **150** may be implemented as various types, such as random access memory (RAM) such as dynamic random access memory (DRAM) and static random access memory (SRAM), read-only memory (ROM), and electrically erasable programmable read-only memory (EEPROM).

As shown in FIG. 3, the portable electronic device **100a** may further include an external memory **160**. The other components of the portable electronic device **100a** are the same as those described with reference to FIG. 2, and thus, descriptions thereof will be omitted herein.

The external memory **160** is electrically connected to a processor **110** and stores data received through a Bluetooth module **120**. The external memory **160** may be connected to the portable electronic device **100** through an additional connector and/or cable. The external memory **160** may further include a component for enabling charging using an

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additional cable even when the external memory **160** is connected to the portable electronic device **100**.

Here, the processor **110** may encode bio-signal data with a preset encryption key and store the encoded bio-signal data in the external memory **160**. Also, the processor **110** may store, in the external memory **160**, an output control signal for outputting the bio-signal data and/or commands for processing of the output control signal. The processor **110** may transmit an encryption key or a decryption key for the bio-signal data before the output control signal.

The processor **110** may be designed to undergo a preset authentication process to access the external memory **160** and data stored in the external memory **160**. For example, the processor **110** may receive the bio-signal data and/or the output control signal from the external memory **160** by inputting a preset password or undergoing a preset login process.

FIG. 4 is a block diagram of an electronic device **200** according to one or more embodiments.

As shown in FIG. 4, the electronic device **200** may include a processor **210**, a communication interface unit **220**, a memory **230**, an output unit **240**, and an input unit **250**.

The processor **210** may be implemented to receive an output control signal of bio-signal data stored in the portable electronic device **100** recognized through the communication interface unit **220** and to output the bio-signal data received via the portable electronic device **100** through the output unit **240** according to the output control signal. The processor **210** may execute an authentication process under control of the portable electronic device **100**. The authentication process may identify a password input through the portable electronic device **100** or the electronic device **200**. The authentication process may identify a stored certificate in the electronic device **200**. Also, the authentication process may be a login process to an external server.

Under control of the portable electronic device **100**, a certificate stored in the memory **230** of the electronic device **200** may be considered to determine whether or not the portable electronic device **100** is approved. When the portable electronic device **100** is determined to be approved, the processor **210** may store the bio-signal data received via the portable electronic device **100** in the memory **230**. When the portable electronic device **100** is determined to be approved, the processor **210** may transmit the bio-signal data to an external cloud server. When the portable electronic device **100** to which the electronic device **200** is connected is determined to be approved, the processor **210** may transmit the bio-signal data to a preset electronic address. The processor **210** may receive, in real time, bio-signal data measured through the bio-signal measuring device **300** and immediately output the bio-signal data through the output unit **240**.

The communication interface unit **220** refers to an interface which communicates in a wired or wireless method. A USB is an example of the wired method of the communication interface unit **220**, and WiFi is an example of the wireless method of the communication interface unit **220**. The communication interface unit **220** may receive the bio-signal data from the portable electronic device **100**.

The memory **230** may temporarily store data received from the portable electronic device **100**. For example, the memory **230** may store the output control signal received from the portable electronic device **100** and delete the executed output control signal after the output is performed.

The memory **230** may permanently store data received from the portable electronic device **100**. Data may be

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permanently stored in the memory **230** according to a control signal of the portable electronic device **100**. The memory **230** may be implemented as various types, such as random access memory (RAM) such as dynamic random access memory (DRAM) and static random access memory (SRAM), read-only memory (ROM), hard disk driver, and electrically erasable programmable read-only memory (EEPROM or flash memory).

The output unit **240** may display a screen for providing information. A user interface generated by the processor **210** may be output. The output unit **240** may provide a screen for outputting the bio-signal data according to the control signal received from the portable electronic device **100**. The output unit **240** may output the authentication process according to a control signal received from the portable electronic device **100**. For example, the output unit **240** may include a display panel or the like for displaying a screen. Also, the output unit **240** may include a display panel, a speaker, or the like.

The input unit **250** may receive an input from a user. The input unit **250** may include an operation panel for receiving a user input. In detail, the input unit **250** may include devices capable of receiving various types of user inputs, such as a keyboard, a physical button, a touch screen, a camera, and a microphone. The input unit **250** may receive a preset user input under control of the portable electronic device **100**. A user input that is input through the input unit **250** may be transmitted to the portable electronic device **100** through the communication interface unit **220**.

The electronic device **200** communicates with the portable electronic device **100** and outputs data received from the bio-signal measuring device **300** or performs an operation under control of the portable electronic device **100**.

FIGS. **5** through **8** are flowcharts according to one or more embodiments.

FIG. **5** is a flowchart of a method of operation of the portable electronic device **100**. In operation **S110**, the portable electronic device **100** is paired with the bio-signal measuring device **300** and receives bio-signal data measured by the bio-signal measuring device **300**.

In operation **S120**, the portable electronic device **100** stores the received bio-signal data in a memory.

In operation **S130**, the portable electronic device **100** may detect or recognize the electronic device **200** through a communication interface unit. The portable electronic device **100** may detect connection of the electronic device **200** through the communication interface unit on the basis of periodical connection detection or an interrupt signal.

In operation **S140**, the portable electronic device **100** determines whether or not the electronic device **200** is approved by receiving preset authentication data from the detected electronic device **200**. The authentication data may be generated on the basis of data stored in the electronic device **200**, code input by a user, or identification information stored in a system area of the electronic device **200**. The portable electronic device **100** may determine whether or not the electronic device **200** is approved by receiving, from the electronic device **200**, various types of authentication data such as a password, an ID and a password, or a stored certificate.

In operation **S150**, the portable electronic device **100** transmits, to the electronic device **200** that was approved, the bio-signal data and/or a control signal related to the bio-signal data. Through the electronic device **200**, which is preregistered or approved, the portable electronic device **100** may output a bio-signal, store the bio-signal, or transmit the bio-signal to another device.

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FIG. **6** is a flowchart of a method of operation of the electronic device **200**, which receives bio-signal data from the portable electronic device **100**.

In operation **S210**, the electronic device **200**, which receives the bio-signal data, detects and communicates with the portable electronic device **100** through a communication interface unit.

In operation **S220**, the electronic device **200**, which receives the bio-signal data, may receive the bio-signal data and store the bio-signal data in a memory under control of the portable electronic device **100**. The electronic device **200** may output a preset user interface to control the portable electronic device **100**.

In operation **S230**, the electronic device **200**, which receives the bio-signal data, may receive an output control signal of the bio-signal data and output the bio-signal data through an output unit under control of the portable electronic device **100**.

In operation **S240**, the electronic device **200**, which receives the bio-signal data, may transmit, to the portable electronic device **100**, a user input that is input by a request of the portable electronic device **100** and execute a control signal received from the portable electronic device **100**.

In operation **S250**, the electronic device **200**, which receives the bio-signal data, may stop processing for a bio-signal when connection of the portable electronic device **100** is removed.

FIG. **7** is a flowchart showing operations between two portable electronic devices **100**, **200** according to one or more embodiments. FIG. **8** is a flowchart showing other operations between two portable electronic devices **200**, **100** according to one or more embodiments;

As shown in FIG. **7**, in operation **S310**, the portable electronic device **100** may detect the electronic device **200** through a communication interface unit. The portable electronic device **100** may establish a communication session for communicating with the detected electronic device **200**.

In operation **S320**, the portable electronic device **100** may determine whether or not the portable electronic device **100** is in a state of receiving bio-signal data while being connected to the electronic device **200**. The portable electronic device **100** may determine whether or not the portable electronic device **100** is in the state of receiving the bio-signal data on the basis of whether or not the portable electronic device **100** is communicating with the bio-signal measuring device **300** and is in a state of receiving data from the bio-signal measuring device **300**.

In operation **S330**, after a time point at which the electronic device **200** is detected, the portable electronic device **100** may transmit, to the electronic device **200**, an output control signal implemented to output the bio-signal data received from the bio-signal measuring device **300** while being connected to the electronic device **200**. In this case, the output control signal may be a signal that controls data received from the bio-signal measuring device **300** to be output in real time or after a preset time (after a very short time).

In operation **S340**, the portable electronic device **100** may output information about the bio-signal data stored in a memory through an output unit of the electronic device **200**.

In operation **S341**, the portable electronic device **100** may receive, through an input unit of the electronic device **200**, an output selection signal including object information and time information. The portable electronic device **100** may provide a screen for selecting the object information and the time information to be output in response to an access signal from the electronic device **200**. The object information may

be information about an object from which bio-signal data is to be measured and may be information about a living body such as a person or an animal. The object information refers to information about an object to be output. The time information may include a start value, an end value, and the like of a particular time at which the bio-signal data is to be output, from among a time interval in which the bio-signal data is recorded.

In operation S342, the portable electronic device 100 may generate the output control signal corresponding to the output selection signal and transmit the output control signal to the electronic device 200. The portable electronic device 100 may specify the object information and the time information included in the output selection signal and extract bio-signal data based on the object information and the time information. The portable electronic device 100 may generate an output control signal including commands output by a user interface outputting the bio-signal data and transmit the output control signal to the electronic device 200. The output control signal may be generated considering the resolution of the electronic device 200.

The electronic device 200 may output an output screen in response to the received output control signal. As a result, measured bio-signal data may be outputted through the electronic device 200.

As shown in FIG. 8, in operation S410, the electronic device 200 may detect the portable electronic device 100.

In operation S420, the electronic device 200 may receive an access signal from the detected portable electronic device 100.

In operation S430, the electronic device 200 may output a screen for inputting an authentication code in response to the access signal.

In operation S440, the electronic device 200 may transmit, to the portable electronic device 100, the input authentication code or an access code generated using a key stored in the electronic device 200. The key stored in the electronic device 200 may include at least one of an identification code, a phone number, and a serial code of the electronic device 200. The authentication code may be acquired through the portable electronic device 100 or another electronic device 200.

In operation S450, when the access code is determined to be true, the electronic device 200 may receive a control signal related to an output or copy of the bio-signal data stored in the portable electronic device 100. The access code may be determined to be true through the portable electronic device 100.

FIG. 9 is a block diagram of an electrocardiogram measuring device 300 according to one or more embodiments.

As shown in FIG. 9, the electrocardiogram measuring device 300 may include a signal detector 310, a signal converter 320, a processor 330, a sensor unit 340, a power supply unit 350, a memory 360, and a communicator 370 to detect an electrical signal generated from the heart of an object.

The signal detector 310 may detect a bio-signal such as blood pressure, blood sugar, heartrate, oxygen saturation level, sleep time, depression diagnosis, chemical analysis, weight, the amount of exercise, or an electrocardiogram. The signal detector 310 outputs an electrocardiogram analog signal by detecting an electrical signal from the heart of an object. The signal converter 320 may convert the bio-signal into a digital signal. The signal converter 320 may convert the electrocardiogram analog signal into an electrocardiogram digital signal.

The processor 330 may be implemented to operate in several operation modes in association with processing of the electrocardiogram digital signal. The processor 330 may operate in one of a first operation mode for storing the electrocardiogram digital signal in the memory 360 and a second operation mode for transmitting the electrocardiogram digital signal to an external electronic device. The processor 330 may operate in the first operation mode or the second operation mode or may operate simultaneously in the first operation mode and the second operation mode.

In the second operation mode, the processor 330 may transmit, to the external electronic device, the electrocardiogram digital signal stored in the memory 360. The external electronic device may be implemented as a smart-phone, a table personal computer (PC), a mobile phone, a video phone, an e-book reader, a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, a camera, a wearable device, or the like, including one or more processors. The external electronic device includes a memory and may be in the form of a storage medium including a processor for controlling the memory. The external electronic device may include a communication interface for communicating with the electrocardiogram measuring device 300 or may be connected to an additional communication interface to operate.

In the first operation mode, the processor 330 may store, in the memory, an electrocardiogram digital signal merged with a user input. The processor 330 may receive the user input from an input unit (not shown), which is electrically connected, or a user terminal, which is connected through a communication network. The processor 330 may convert an electrocardiogram signal by using the received user input and generation time information of the user input.

In the first operation mode, the processor 330 may merge the electrocardiogram digital signal with the user input and store, in the memory 360, the electrocardiogram digital signal merged with the user input by using the generation time information of the user input.

In the second operation mode, the processor 330 may transmit, to the external electronic device, the electrocardiogram digital signal merged with the user input. The external electronic device may store the received electrocardiogram digital signal in a memory or may output the received electrocardiogram digital signal through an output unit.

The processor 330 may automatically change an operation mode considering a communication state with the external electronic device. For example, when the communication state with the external electronic device is abnormal, the processor 330 may stop the second operation mode and change the second operation mode to the first operation mode to operate.

When the communication state is normal again, the operation mode is changed to the second operation mode again. After the operation mode is changed to the second operation mode, an electrocardiogram digital signal measured in real time may be transmitted to the external electronic device.

The processor 330 may further include a temperature sensor, a motion sensor, a proximity sensor, a sound sensor, or the like.

The processor 330 may generate an electrocardiogram digital signal merged with a tag corresponding to information acquired by analyzing the electrocardiogram digital signal. The processor 330 may operate in the first operation

mode and/or the second operation mode for the electrocardiogram digital signal merged with the tag.

The information acquired by analyzing the electrocardiogram digital signal may be a heartrate, cardiac component information, the maximum magnitude value, time information of a cardiac feature point, waveform magnitude information of the cardiac feature point, or the like. The maximum magnitude value refers to the maximum value of the magnitude from among the signal magnitudes of the electrocardiogram digital signal. The cardiac feature point refers to a point at which a slope of a differential value of the electrocardiogram digital signal changes from positive to negative or negative to positive. The processor **330** may store, in the memory **360**, a tag including at least one of the heartrate, the maximum magnitude value, the time information of the cardiac feature point, and the waveform magnitude information of the cardiac feature point acquired by analyzing the electrocardiogram digital signal, in association with the electrocardiogram digital signal.

The processor **330** may acquire the amount of exercise of an object, further generate a tag corresponding to the exercise amount, and merge the tag corresponding to the exercise amount with the electrocardiogram digital signal.

In the second operation mode, the processor **330** may transmit in real time, to the external electronic device, the electrocardiogram digital signal merged with the user input or the tag. Alternatively, when the measurement of the electrocardiogram digital signal is completed, the processor **330** may transmit, to the external electronic device **100, 200**, the electrocardiogram digital signal stored in the memory **360**. When the measurement of the electrocardiogram digital signal is completed, the processor **330** may generate a measurement completion signal and start uploading, to the external electronic device, the electrocardiogram digital signal stored in the memory **360** by the measurement completion signal. The measurement completion signal may be generated by an input from the external electronic device, an input from an input unit (not shown), or a preset measurement time, or may be generated after a preset measurement time from a measurement start time.

The power supply unit **350** supplies power to the signal detector **310**, the signal converter **320**, the processor **330**, the sensor unit **340**, the memory **360**, the communicator **370**, and the like. The power supply unit **350** may use a rechargeable battery or a disposable battery.

The memory **360** may store data related to measurement of an electrocardiogram. The memory **360** may store information about the object, a measurement time, and data related to a measured electrocardiogram. The memory **360** may be hardware storing various types of data processed in the electrocardiogram measuring device **300** and may store pieces of data processed by the processor **330** and pieces of data to be processed by the processor **330**. The memory **360** may store a generated input and a tag related to the input. The memory **360** may store an electrocardiogram digital signal and an electrocardiogram digital signal synchronized with a tag. The memory **360** may be implemented as various types, such as random access memory (RAM) such as dynamic random access memory (DRAM) and static random access memory (SRAM), read-only memory (ROM), and electrically erasable programmable read-only memory (EEPROM or flash memory).

The sensor unit **340** senses temperature or movement. The sensor unit **340** may sense movements in x, y, and z-axis directions, angular movement, and roll-pitch-yaw rotational movement. The sensor unit **340** may sense the momentum of a device, a temperature value of the device, physical pres-

sure applied to the device, and the like. The sensor unit **340** may include an element for sensing temperature, an element for sensing motion, an element of sensing pressure, and the like.

The communicator **370** may receive a control signal from an external user terminal. The communicator **370** may transmit measured data to the external user terminal.

The device described above may be implemented with hardware components, software components, and/or combinations of hardware components and software components. For example, the apparatus and components described in one or more embodiments may be implemented, for example, by using one or more general-purpose computers or special-purpose computers such as a processor, a controller, an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a programmable logic unit (PLU), a microprocessor, or any other devices capable of executing and responding to instructions. A processing unit may perform an operating system (OS) and one or more software applications executed on the OS. Also, the processing unit may access, store, control, process, and generate data in response to execution of software. For convenience of understanding, while one processing unit has been described as being used, it will be understood by one of ordinary skill in the art that the processing unit may include a plurality of processing elements and/or a plurality of types of processing elements. For example, the processing unit may include a plurality of processors or one processor and one controller. Also, the processing unit may include another processing configuration such as a parallel processor.

Software may include a computer program, code, instructions, or a combination of one or more thereof and may configure the processing unit to operate as wanted or instruct the processing unit independently or collectively. Software and/or data may be permanently or temporarily embodied in any type of machine, component, physical device, virtual equipment, computer storage medium or device, or transmitted signal wave to be interpreted by the processing unit or to provide the processing unit with instructions or data. Software may be distributed over networked computer systems and may be stored or executed in a distributed manner. Software and/or data may be stored on one or more computer-readable recording media.

The method according to one or more embodiments may be implemented in the form of a computer instruction that may be executed through various types of computer units and recorded on a computer-readable recording medium. The computer-readable recording medium may include program instructions, data files, data structures, or the like alone or in combination. The program instructions recorded on the medium may be specially designed and configured for an embodiment or may be known and available to one of ordinary skill in computer software. Examples of the computer-readable recording medium include magnetic media such as hard disks, floppy disks, and magnetic tapes, optical media such as a CD-ROM and a DVD, magneto-optical media such as floptical disks, and hardware devices specially configured to store and perform program instructions such as ROM, RAM, and flash memory. Examples of the program instructions include high-level language code that may be executed by a computer using an interpreter or the like, as well as machine language code made by a compiler. The hardware device described above may be configured to operate as one or more software modules to perform operations of the embodiments, and the reverse is the same.

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According to one embodiment as described above, a portable electronic device may receive bio-signal data from a measuring device, and store and carry the bio-signal data.

Also, bio-signal data stored in the portable electronic device or bio-signal data received through the portable electronic device may be provided through an output unit of an electronic device connected to the portable electronic device.

It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the following claims.

What is claimed is:

1. An electrocardiogram measuring system comprising: an electrocardiogram measuring device comprising:
 - a signal detector detecting an electrocardiogram analog signal;
 - a signal converter converting the electrocardiogram analog signal into an electrocardiogram digital signal;
 - a memory;
 - a communicator communicating with an external device; and
 - a processor configured to cause the electrocardiogram measuring device to operate in a first operation mode for storing the electrocardiogram digital signal in the memory and a second operation mode for transmitting the electrocardiogram digital signal to the external device; and
 a portable electronic device communicatively coupled to the electrocardiogram measuring device and the external device, the portable electronic device configured to facilitate the second operation mode of transmitting the electrocardiogram digital signal from the electrocardiogram measuring device to the external device by pre-storing an identification (ID) of the external device, accessing a certificate stored in the external device, and authenticating the certificate of the external device, the portable electronic device further configured to transmit the electrocardiogram digital signal to the external device to store the electrocardiogram digital signal in the external device.
2. The electrocardiogram measuring system of claim 1, wherein the processor is configured to transmit the electrocardiogram digital signal to a cloud server.
3. The electrocardiogram measuring system of claim 1, wherein the portable electronic device is configured to select the external device by a program.
4. The electrocardiogram measuring system of claim 3, wherein the program is stored in the portable electronic device.
5. The electrocardiogram measuring system of claim 1, wherein the electrocardiogram digital signal is transmitted to the portable electronic device and generated on an output display of the portable electronic device before the portable electronic device connect to the external device.

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6. The electrocardiogram measuring system of claim 1, wherein the electrocardiogram digital signal is transmitted to the external device and generated on an output display of the external device while connected to the electrocardiogram measuring device.

7. The electrocardiogram measuring system of claim 1, wherein the portable electronic device is configured to run authenticating the certificate of the external device by user input while the electrocardiogram digital signal is output on an output unit of the portable electronic device.

8. The electrocardiogram measuring system of claim 1, wherein the portable electronic device is configured to transmit the electrocardiogram digital signal to one or more device.

9. The electrocardiogram measuring system of claim 1, wherein the portable electronic device further comprises an external memory, and the external memory stores the electrocardiogram digital signal.

10. An electrocardiogram measuring method comprising: communicating, by a portable electronic device including a processor, with an electrocardiogram measuring device and an external device,

wherein the electrocardiogram measuring device is communicatively coupled to the external device and configured to detect an electrocardiogram analog signal, convert the electrocardiogram analog signal into an electrocardiogram digital signal, and set an operation mode to a first operation mode for storing the electrocardiogram digital signal in a memory or a second operation mode for transmitting the electrocardiogram digital signal to an external device;

facilitating, by the portable electronic device, the second operation mode of transmitting the electrocardiogram digital signal from the electrocardiogram measuring device to the external device by pre-storing an identification (ID) of the external device, accessing a certificate stored in the external device, and authenticating the external device; and

transmitting, by the portable electronic device, the electrocardiogram digital signal to the external device so that the electrocardiogram digital signal is stored in the external device.

11. The electrocardiogram measuring method of claim 10, further comprising:

while the portable electronic device is running, displaying the electrocardiogram digital signal by the portable electronic device.

12. The electrocardiogram measuring method of claim 10, further comprising: storing, by the portable electronic device, the electrocardiogram digital signal in the memory or a cloud server through a network.

13. The electrocardiogram measuring method of claim 11, further comprising:

controlling, by the portable electronic device, a user interface for outputting the electrocardiogram digital signal and object information to be displayed at the external device while connected to the electrocardiogram measuring device.

14. The electrocardiogram measuring method of claim 10, further comprising controlling, by the portable electronic device, the electrocardiogram digital signal to be displayed at the external device while connected to the electrocardiogram measuring device.

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